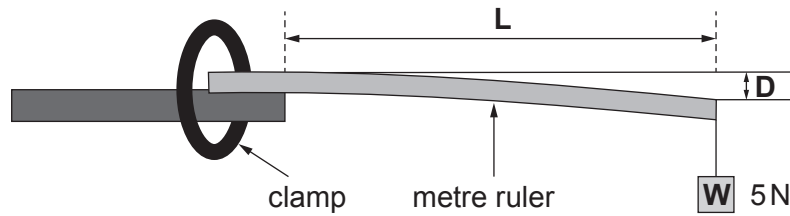


Answer **all** questions.

1. Students are investigating how the stiffness of a wooden metre ruler depends on its length. They use the apparatus shown below. The weight, **W**, on the end of the ruler remains constant at 5 N.



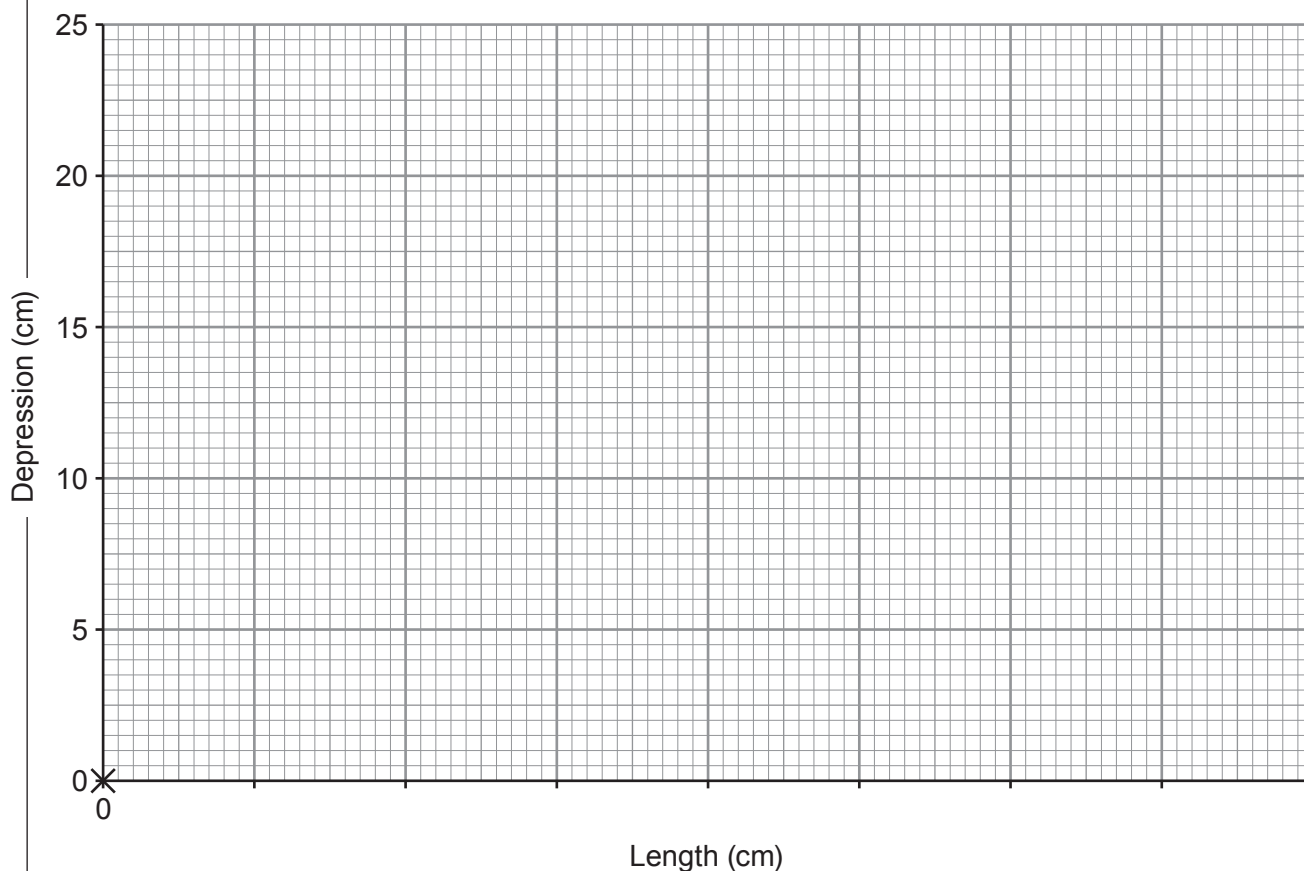
The length, **L**, is varied. For each length, **L**, the depression, **D**, of the end of the ruler is measured.

Their results are shown in the table below.

Length L (cm)	Depression D (cm)
0	0.0
30	1.0
40	3.0
50	6.0
60	10.0
70	15.0
80	22.0



- (a) Plot the data on the grid below and draw a smooth curve. The point (0,0) has already been plotted for you. [4]



- (b) It was expected that depression, **D**, would be directly proportional to length, **L**. Explain whether the graph shows a proportional relationship. [2]

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.....

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- (c) Use information in the table and the equation:

$$\text{stiffness} = \frac{\text{weight}}{\text{depression}}$$

to calculate the stiffness of the ruler at a length of 50 cm. [3]

stiffness = N/cm



- (c) Many countries have banned the use of genetically modified foods.

Suggest **two** reasons for their concern.

[2]

1.

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2.

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- (d) The home brewer also uses this genetically modified yeast to produce beer. Complete the steps in the fermentation process. Three steps have been given. [4]

Step 1: Barley is mixed with water. Starch in the barley is converted to glucose.

Step 2: Hops are added for flavour.

Step 3:

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Step 4:

.....

Step 5:

.....

Step 6:

.....

Step 7: The beer may be pasteurised, then stored in bottles or casks.

